

BASIC FORMULAE USED IN PRIMARY MATHEMATICS

1. PERIMETER

Perimeter of a rectangle = $L + W + L + W$ Or $2(L + W)$

Perimeter of a triangle = $S + S + S$

Perimeter of a square = $S + S + S + S$ Or $4S$

Perimeter of the rhombus = $S + S + S + S$ Or $4S$

Perimeter of any regular polygon = No of sides X length.

Circumference of a circle = $2\pi r$ or πd

Circumference of a semi circumference = $\frac{1}{2} \pi d$

Circumference of a quadrant = $\frac{1}{4} \pi d$

Circumference of a sector = $\frac{\theta}{360^\circ} \pi d$ where θ = angle sector.

Perimeter of a semi-circle = $\frac{1}{2} \pi d + d$

Perimeter of a quadrant = $\frac{1}{4} \times 2\pi r + r + r$

2. AREA (Expressed in square units)

Area of a rectangle = $L \times W$

Area of a square = $S \times S$ or S^2

Area of a triangle = $\frac{1}{2} b \times h$ or $\frac{b \times h}{2}$

Area of a trapezium = $\frac{1}{2} h(a + b)$ or $\frac{h(a+b)}{2}$

Area of a parallelogram = $b \times h$

Area of a rhombus = $\frac{1}{2} \times d_1 \times d_2$ or $\frac{1}{2} bh \times 4$ right angled triangles

Area of a kite = $\frac{1}{2} \times d_1 \times d_2$ or $\frac{1}{2} bh \times 4$ right angled triangles

Area of a circle = πr^2

Area of a sector = $\frac{\theta}{360^\circ} \pi r^2$ where θ = angle sector.

Area of a semi-circle = $\frac{1}{2} \pi r^2$

Area of a quadrant = $\frac{1}{4} \pi r^2$

3. TOTAL SURFACE AREA

T.S.A of cube = $6s^2$

T.S.A of a cylinder = $2(L \times W) + 2(W \times H) + 2(L \times H)$

T.S.A of closed cylinder = $2\pi r^2 + 2\pi rh$

T.S.A of open cylinder = $\pi r^2 + 2\pi rh$

T.S.A of a sphere = $4\pi r^2$

T.S.A of a triangular prism = $2\left(\frac{1}{2} b \times h\right) + (L \times W) + (L \times W) + (L \times W)$

4. **VOLUME (expressed in cubic units)**

Volume of a cube = base area X height

$$= S \times S \times S$$

$$= S^3$$

Volume of a cuboid = base area X height.

$$= (L \times W) \times H$$

Volume of a triangular prism = base area X length

$$= \left(\frac{1}{2} \times b \times h\right) \times L$$

Volume of a cylinder = base area X height

$$= (\pi r^2) \times h$$

Volume of a sphere = $\frac{4}{3} \pi r^3$

Volume of a cone = $\frac{1}{3} \pi r^2 h$

5. **MEASURES**

Distance = Speed X Time (km or m)

Time = $\frac{\text{distance}}{\text{speed}}$ (hours)

Speed = $\frac{\text{distance}}{\text{time}}$ (km/hr)

To expressed Km/hr to m/s = $\frac{d \times 1000}{60 \times 60}$ (m/s) where *d* stands for distance given

To expressed m/s to km/hr = $\frac{d \times 60 \times 60}{1000}$ (m/s) where *d* stands for distance given

Capacity = $\frac{\text{volume}}{1000 \text{ cm}^3}$ (litres)

Profit = selling price - cost price

Loss = cost price - selling price

Selling = cost price + profit (when given profit)

Cost price = selling price - loss (when given loss)

Discount = cash paid - buying price

Percentage profit = $\left(\frac{\text{profit}}{\text{cost price}} \times 100\right) \%$

Percentage loss = $\left(\frac{\text{loss}}{\text{cost price}} \times 100\right) \%$

Percentage discount = $\left(\frac{\text{discount}}{\text{marked price}} \times 100\right) \%$

Simple interest = Principal X Rate X Time

Amount = Principal + Interest

Mean(average) = $\frac{\text{sum of items}}{\text{number of items}}$

Sum of items when given Average and number = Average X Number of items.

TRIANGULATION

Number of right angles = $2n - 4$ where n stands for sides

Number of triangles = $(n - 2)$ where n stands for sides

Interior angle sum = $180^\circ(n - 2)$ or $90^\circ(2n - 4)$ where n stands for sides

Exterior angles = $\frac{360^\circ}{\text{number of sides}}$

Number of sides of a regular polygon = $\frac{360^\circ}{\text{Each exterior angle}}$

Number of revolutions = $\frac{\text{distance covered}}{\text{circumference}}$

FACT SUMMARY/EQUIVALENCIES

1km = 1000m

$1m^3$ = 1000litres

1hectare = $10,000m^2$

1 tonne = 1000kg

$1m^2$ = $10,000m^2$

1litre = 1000ml

1 day = 24 hours

1 fortnight = 14days

A score = 20 items

1dozen = 12 items

A gross = 144 items

1 hour = 60mins

1hour = 3600sec

Millennium = 1000years

Century = 100years

Golden jubilee = 50years

Silver jubilee = 25years

Decade = 10 years

Leap year = 366years

Ordinary year = 365years